

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	28 June 2026
Team ID	
Project Name	Metro Ticket Generation System Using ServiceNow
Maximum Marks	4 Marks

Technical Architecture:

The Metro Ticket Generation System is built entirely on the ServiceNow platform. The architecture follows a layered design with a clear separation of Presentation, Application Logic, Workflow, Data, and Administration layers. All components reside within ServiceNow's cloud environment, eliminating the need for external servers or infrastructure.

Architecture Flow: Commuter (Web Browser) → ServiceNow Service Portal → Service Catalog (Metro Ticket Booking Item) → Client Scripts / UI Policies → Form Submission → Flow Designer → Custom Metro Booking Table → Business Rules → Admin Dashboard

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Commuter interacts via a Service Catalog form on the ServiceNow Service Portal	ServiceNow Service Portal, Service Catalog, Form Designer
2	Application Logic – Validation	Validates mandatory form fields before submission	Client Scripts (ServiceNow)
3	Application Logic – Fare Calc	Automatically calculates fare based on selected route and passenger count	Client Scripts (ServiceNow)
4	UI Control	Controls field visibility and mandatory status dynamically on the booking form	UI Policies (ServiceNow)
5	Workflow Automation	Captures catalog variables and automates booking record creation on form submission	Flow Designer (ServiceNow)
6	Backend Logic	Executes server-side logic for data consistency and status updates	Business Rules (ServiceNow)

7	Database	Stores all metro ticket booking records including route, fare, passenger info, and status	ServiceNow Custom Table (Metro Booking Table)
8	Administration	Admins manage and track all booking records	ServiceNow List View / Admin Dashboard
9	Infrastructure / Cloud	Application hosted on ServiceNow cloud — no external servers required	ServiceNow Cloud Platform (SaaS)

Table 2: Application Characteristics

S.No	Characteristic	Description	Technology
1	Platform Frameworks	ServiceNow platform provides built-in frameworks for UI, workflow, and data management	ServiceNow Application Framework
2	Security	Role-based access control (RBAC) for commuters and administrators; ServiceNow ACL rules protect data	ServiceNow ACLs, RBAC
3	Scalable Architecture	Modular design with separate catalog item, workflow, and database — easily extensible	ServiceNow Modular Architecture
4	Availability	ServiceNow cloud guarantees high availability with minimal downtime via redundant infrastructure	ServiceNow Cloud SLA
5	Performance	Client Scripts execute in-browser for instant form validation; Flow Designer processes async for efficiency	Client Scripts, Flow Designer